

RapidBoxForest: Certificate-Aware Box Reuse for Multi-Query Manipulation

Anonymous Authors

Abstract—Multi-query manipulation in semi-static workcells benefits from reusable free-space structure, but existing reuse objects tend to be either edge-centric roadmaps or expensive convex-region covers. RapidBoxForest (RBF) introduces a middle ground: a reusable forest of axis-aligned configuration-space (C-space) boxes validated by endpoint-to-link swept-capsule envelopes. RBF separates kinematics evidence from scene labels through a Lifelong Envelope Cache Tree (LECT), recomputes obstacle labels for each scene, and uses a typed corridor graph to distinguish certificate-backed regions from paths that require fixed-step validation. Under saved Shelf+IIWA catalogs and a common 8-thread policy, a prebuilt depth-23 LECT cache gives 0.171 s median reusable build time and 0.077 s median per-query online time, versus 2.770 s live HIFK-5 build time. Across random-scene catalogs on three robot models, RBF trades some RRT-Connect per-query speed for shorter validated paths and lower online time than BIT*. Local edit experiments report maintenance-only cache-update speedups of 153.3–207.5 $\times$ for insertion and 943.6–3354.0 $\times$ for removal over a 15-obstacle warm rebuild.

Index Terms—Motion planning, manipulation planning, multi-query planning, collision checking, configuration-space decomposition, link envelopes.

I. INTRODUCTION

Many high-degree-of-freedom manipulation systems solve streams of related queries in nearly fixed workcells. Robot kinematics persist, while start-goal pairs, local obstacles, and small scene edits change. This pattern appears in machine tending, shelf transfer, inspection, and pick-and-place cells, where the robot model is fixed but clutter and requested endpoints vary across a shift. A useful reusable object for this setting should therefore do more than store a previous path: it should accelerate later queries, remain cheap to update, and state clearly which parts of a returned path are certified.

Existing reuse mechanisms tend to emphasize one side of this trade-off. Sampling-based planners and roadmap variants reuse vertices, edges, or prior paths [4]–[7], but the reused object is usually non-volumetric. Convex-region methods such as Iterative Regional Inflation (IRIS) and Graphs of Convex Sets (GCS) provide optimization-ready free-space regions [1]–[3], but their regions can be expensive to construct, certify, and repair after edits.

RapidBoxForest (RBF) targets the middle ground. It grows reusable axis-aligned configuration-space (C-space) boxes around scene-relevant seeds, validates them with endpoint-to-link swept-capsule envelopes, and connects accepted boxes in a typed corridor graph. The boxes are not the tightest possible C-space regions; they are chosen because replay, adjacency, invalidation, and local repair remain simple enough to expose directly.

a) *Problem statement*: Let $\mathcal{Q} \subset \mathbb{R}^n$ be the joint-limit box of a fixed n -degree-of-freedom manipulator and let $\mathcal{O} \subset \mathbb{R}^3$ be the current workspace obstacle set. Given a set or stream of start-goal queries $(q_s, q_g) \in \mathcal{Q}^2$, RBF constructs a finite family of C-space boxes that pass the configured scene-validation policy and uses those boxes as vertices of an adjacency graph. The design target is a query-agnostic, volumetric, and auditable scene structure, not an exhaustive decomposition of $\mathcal{C}_{\text{free}}$.

b) *Key idea and terminology preview*: The geometric identity behind the method is classical: if the endpoint sweeps of a rounded link are bounded, then the link sweep is bounded by the convex hull of those endpoint bounds plus the link radius. The contribution is to use this identity as a reusable C-space box validator. Three semantic layers are used throughout the paper: *geometry evidence* denotes obstacle-independent endpoint and link envelopes stored in LECT; *scene structure* denotes the current boxes, labels, and typed graph edges recomputed after obstacle checks; and *returned-path status* is certificate-backed only inside conservative regions and otherwise validation-required. The unified guarantee matrix appears in Table II after the collision model is defined.

This paper makes three contributions.

- 1) A *conservative box-region certificate primitive*. Endpoint-to-link swept-capsule bounds convert endpoint envelopes into link envelopes over a C-space box, giving a reusable collision-free box certificate when the endpoint source, radius padding, and separation tests are conservative. This gives the planner a reusable free-volume unit rather than only checked edges.
- 2) A *certificate-aware reusable box-forest planner*. RBF grows policy-accepted C-space boxes, assembles them into a scene forest and typed corridor graph, and exposes whether a returned segment is certificate-backed or validation-required. This lets the query interface make returned-path semantics explicit instead of hiding them inside post hoc checks.

TABLE I
REUSABLE STATE IN PLANNING PARADIGMS. RBF IS POSITIONED AS LOW-OVERHEAD VOLUMETRIC REUSE WITH EXPLICIT RETURNED-PATH SEMANTICS.

Paradigm family	Reused state	Vol.	Update or validation style	Multi-query implication
Roadmap reuse (PRM/LazyPRM)	Vertices, edges, and cached edge checks	No	Recheck affected edges after scene changes	Strong graph reuse, but no reusable free-volume unit.
Tree and anytime sampling (RRT, RRT-Connect, BIT*, FMT*)	Query tree or batch sample graph	No	Primarily query-local; rebuilt or checkpointed	Fast discovery or quality improvement without persistent volumetric scene structure.
Convex-region optimization (IRIS/GCS)	Convex cells and region graph	Yes	Inflate, certify, or recertify regions after edits	Optimization-ready regions, but update cost can dominate short query batches.
RBF	Box forest, typed graph, and LECT evidence	Yes	Recheck boxes, replay kinematics evidence, and validate non-certificate outputs	Low-overhead free-volume reuse with auditable certificate boundaries.

3) *A two-level reuse and maintenance architecture.* LECT indexes robot-kinematic envelope records separately from scene labels; the scene cache handles local obstacle insertions and deletions by rechecking affected boxes and replaying compatible geometry evidence. This separates kinematic reuse from scene relabeling, which makes local edits cheaper and easier to audit.

The experiments are therefore organized around these semantics: reusable build cost, online query time, validation responsibility, and edit maintenance are reported separately.

Fig. 1 summarizes the method framework: link-interval computation produces reusable geometry evidence, RBF growth recomputes scene labels and graph connectivity for the current obstacles, and LECT replay changes construction cost without changing validation responsibility. Sections III–VI make these blocks precise and then evaluate them.

II. RELATED WORK

Region-based planners make collision-free volume an explicit planning object. IRIS variants inflate collision-free convex regions, and Graphs of Convex Sets (GCS) optimize over the induced region graph [1]–[3], [8]–[10]. RBF shares the goal of reusing free volume but uses axis-aligned C-space boxes rather than arbitrary convex cells, explicitly trading expressivity for cheaper local validation, adjacency, cache replay, and maintenance. Table I is therefore not only a taxonomy: it identifies the gap targeted here, namely low-overhead volumetric reuse between roadmap reuse and optimization-ready convex covers.

Synthesis. Existing reuse objects are either cheap but edge-centric, or volumetric but costly to construct and repair. RBF is not intended to dominate either class universally. Its intended niche is repeated manipulation in scenes where a volumetric reusable layer is valuable, but repeatedly inflating optimization-ready convex regions or rebuilding large roadmaps is too expensive for local edits and short query batches.

a) Continuous-collision-detection endpoint bounds: Swept-volume and interval-kinematics methods supply the geometric primitives used here. Convex-hull generator bounds and capsule radius lifts provide swept-set

enclosures, while the Proximity Query Package (PQP), Gilbert–Johnson–Keerthi (GJK), the Flexible Collision Library (FCL), and interval Denavit–Hartenberg propagation provide collision-query and interval-kinematics primitives used in related continuous-collision-detection pipelines [11]–[22]. Redon *et al.* combine such primitives for articulated-body continuous collision detection over time steps [23], [24]; related Taylor-model and safety methods use similar swept-sphere bounds [25]–[27]. The novelty in RBF is not the endpoint bound alone, but its use as a reusable joint-box validator: compatible evidence is stored in LECT, replayed before scene-specific obstacle tests, and interpreted through the claim boundary in Table II.

b) Graph reuse: Reusable graph planners amortize effort across related queries. Probabilistic roadmaps (PRMs) and LazyPRM reuse roadmaps and edge checks [6], [28]–[30], experience-based systems reuse prior paths [31], and adaptive cell methods split cells until they obtain a searchable accepted-cell graph [32], [33]. RBF keeps the multi-query graph-search benefit but changes the reused unit: vertices are accepted free-volume boxes, edges are typed by certificate status, and kinematics evidence is separated from scene labels rather than saved as stale obstacle outcomes.

c) Sampling and refinement: Rapidly-exploring random tree (RRT) variants, Batch Informed Trees (BIT*), Fast Marching Tree (FMT*), and the Open Motion Planning Library (OMPL) provide standard single-query planners, anytime references, and benchmarking infrastructure [7], [34]–[39]. Trajectory optimizers refine supplied paths [40]–[43]. RBF is therefore not positioned as a single-query optimal planner; it supplies reusable volumetric context before repair, shortcutting, optimization, and final fixed-step validation.

III. LINK INTERVAL ENVELOPES

This section defines the geometric certificate primitive used later. The link-envelope validation layer maps joint regions to endpoint and link bounds; graph connectivity and query acceptance are handled by later layers. For a joint region $I \subseteq \mathcal{Q}$, endpoint envelopes $E_k(I)$ bound

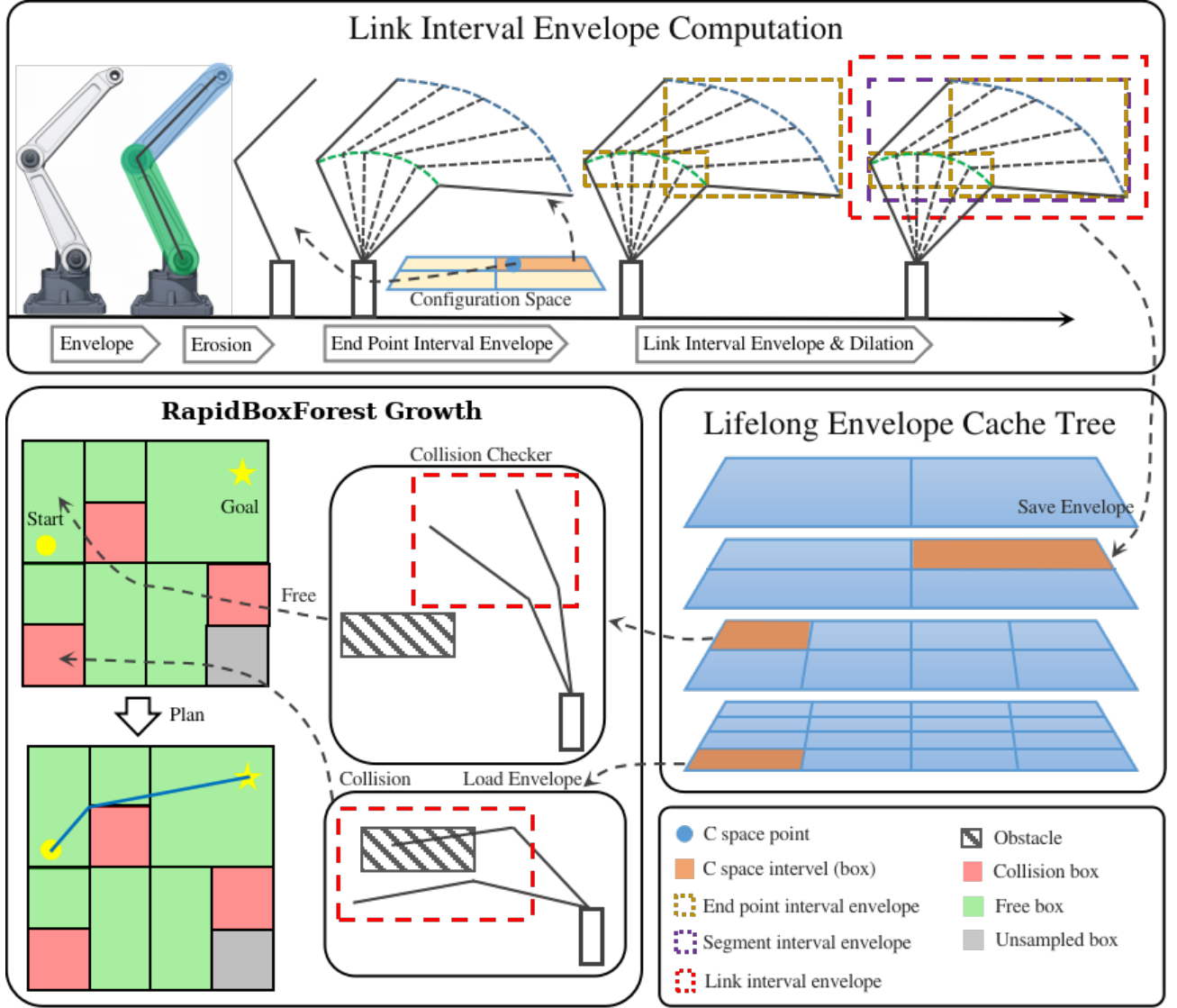


Fig. 1. RBF pipeline. Top: endpoint and segment interval envelopes induce link interval envelopes over a C-space interval. Lower left: sample-guided RBF growth labels free, collision, and unsampled boxes. Lower right: LECT stores kinematics-keyed envelope records for replay. Reused: geometry evidence. Recomputed: obstacle labels and graph edges. Accepted outputs: conservative corridors or paths accepted after fixed-step validation.

endpoint trajectories for endpoint k , and link envelopes $B_\ell(I)$ bound rounded links for link ℓ over the same region:

$$I \mapsto \{E_k(I)\} \mapsto \{B_\ell(I)\}.$$

Axis-aligned C-space boxes are the default regions; local oriented bounding boxes (OBBs) in scaled joint coordinates appear only as typed bridge corridors. Endpoint evidence becomes a certificate only after it is lifted to link envelopes and those envelopes pass conservative scene-separation tests. All empirical or otherwise non-conservative endpoint sources are treated as filters under Table II.

Assumption 1 (Collision model for certificates). *The certificate model covers only workspace-obstacle collision between active rounded links and $\mathcal{O}$; joint limits are enforced separately by requiring $I \subseteq \mathcal{Q}$. Self-collision, attachments, grippers, task regions, and other constraints are outside this certificate model unless encoded in the validation policy and cache descriptor. The experiments instantiate this subset with active-link versus workspace-obstacle collision checks.*

Reporting convention. A certificate-backed object is collision-free under Assumption 1; a validation-required object may guide planning but becomes an accepted path only after the fixed-step validation protocol in Section VI. Cache hits change construction cost, not validation responsibility. Planning aids that are not certificate-backed follow the semantics in Table II. Shortcutting or simplification inherits a certificate only when containment inside the certified region is preserved; otherwise it follows

TABLE II
GUARANTEE MATRIX FOR RETURNED OBJECTS. STATUS IS INTERPRETED UNDER ASSUMPTION 1 AND THE EXPERIMENT PROTOCOL IN SECTION VI.

Object or state	Status	Condition
Path inside conservative box union	Certificate-backed	Every traversed box passes conservative endpoint-to-link validation against the current obstacles.
Path inside finite-cover OBB corridor	Certificate-backed	Every cover tile in the local C-space OBB corridor passes the same box test.
Gap, segment, or repair witness	Validation-required	Certificate-backed only after conversion to a conservative box or OBB corridor.
Filter-derived path	Validation-required	Empirical or non-conservative evidence may guide planning but cannot certify coupled extrema.
LECT replay record	No path claim	Reuses descriptor-compatible geometry evidence only; scene labels and graph edges are recomputed.

the validation-required semantics in Table II. Timing claims using Online/q refer to planner online time rather than total accepted-path wall clock unless an additional validation or simplification cost is explicitly reported. Compact tables without audit-cost columns therefore do not support total-latency comparisons.

A. Geometric Foundation

Let C_0 be a body-frame shape. The rigid transform $T(q)x = R(q)x + t(q)$ maps a body-frame point x to workspace at configuration q , with rotation $R(q)$ and translation $t(q)$, and $S_I(C_0)$ denotes the workspace set swept by C_0 as q ranges over I . Let $\mathbb{B}_p(r)$ denote the radius- r ball in the p -norm, and let $\oplus$ denote Minkowski sum. For a capsule $C = S \oplus \mathbb{B}_2(r)$ with zero-radius center segment S , the fixed-radius ball commutes with rigid motion:

$$S_I(C) = S_I(S) \oplus \mathbb{B}_2(r).$$

Thus a rounded link can be bounded by first bounding the zero-radius center segment and then lifting by the link radius. For a convex body $P = \text{conv}(v_1, \dots, v_m)$ with vertices v_i , outer bounds on the vertex trajectories also bound the swept body,

$$S_I(P) \subseteq \text{conv}\left(\bigcup_{i=1}^m \hat{\Gamma}_i(I)\right).$$

Here $\hat{\Gamma}_i(I)$ denotes a conservative outer bound on the workspace trajectory of vertex v_i , $i = 1, \dots, m$, over I . For a capsule link, only the two endpoint trajectories are needed. Let $\Gamma_a(I)$ and $\Gamma_b(I)$ denote the exact workspace trajectories of the proximal and distal endpoints over I . RBF uses these swept-sphere bounds over a configuration-space box domain and stores the resulting bounds as reusable kinematics-keyed bound records.

Proposition 1 (Endpoint envelopes induce link interval envelopes). *Consider a rounded link $L = [a, b] \oplus \mathbb{B}_2(r)$ and a joint interval I . If endpoint envelopes $E_a(I)$ and $E_b(I)$ satisfy $\Gamma_a(I) \subseteq E_a(I)$ and $\Gamma_b(I) \subseteq E_b(I)$, then*

$$S_I(L) \subseteq \text{conv}(E_a(I) \cup E_b(I)) \oplus \mathbb{B}_2(r).$$

Consequently, any conservative outer representation of the right-hand side is a valid link interval envelope for the whole joint interval I .

Proof. For each $q \in I$, the center segment of the moved link is the convex hull of the two moved endpoints, and those endpoints lie in $\Gamma_a(I)$ and $\Gamma_b(I)$. Hence all center segments are contained in $\text{conv}(\Gamma_a(I) \cup \Gamma_b(I))$. The endpoint inclusions place this set inside $\text{conv}(E_a(I) \cup E_b(I))$. Adding the fixed radius ball commutes with the rigid-motion sweep of the capsule, giving the stated containment. $\square$

Corollary 1 (Conservative configuration-space box validation). *Let $I \subseteq \mathcal{Q}$ be an axis-aligned configuration-space box and let $\{B_\ell(I)\}$ be conservative link interval envelopes for all active rounded links. If $B_\ell(I) \cap \mathcal{O} = \emptyset$ for every link ℓ , then every configuration $q \in I$ is collision-free with respect to the collision model in Assumption 1.*

Proof. By Proposition 1, each conservative link envelope contains the swept workspace volume of its link over all configurations in I . This disjointness from $\mathcal{O}$ for every active link excludes workspace obstacle collision for the whole manipulator at every $q \in I$ under Assumption 1. $\square$

a) *Numerical conservatism:* A certificate is claimed only when endpoint propagation, radius padding, and obstacle separation produce outward bounds under the collision model of Assumption 1. For link radius r_ℓ , the axis-aligned bounding box (AABB) broadphase pads by the infinity-norm ball $\mathbb{B}_\infty(r_\ell)$, which conservatively contains the Euclidean radius lift $\mathbb{B}_2(r_\ell)$. SupportHull/GJK denotes a support-function hull with GJK refinement; it is certificate-backed only with outward obstacle inflation, explicit margins, and conservative termination tolerances. Empirical endpoint sources and post-processing outside a conservative box union or a finite-cover OBB-corridor union are treated as non-conservative filters; they do not extend Corollary 1 or Lemma 1.

B. Endpoint Interval Envelopes

For endpoint k , let $\Gamma_k(I)$ be its exact trajectory over joint interval I . Any conservative set containing $\Gamma_k(I)$ is an endpoint interval envelope; the basic record is an endpoint axis-aligned bounding box (AABB) because it is compact and directly reusable by all link-envelope representations. Two certificate-backed endpoint sources are used. Interval forward kinematics (IFK) denotes the unsplit interval

pass; IFK-AA is the affine-arithmetic implementation that propagates bounds along the Denavit–Hartenberg chain, with shared noise symbols preserving part of the trigonometric correlation. Hierarchical IFK (HIFK) bisects the joint interval and forms hulls of child endpoint boxes, reducing wrapping at higher materialization cost. HIFK is a hierarchical split-and-hull evaluation (for example, HIFK-5 means five split levels), so the split depth is part of the cache descriptor and cannot be exchanged with an unsplit record. A critical-sample source probes interval endpoints, midpoints, near-boundary configurations, and trigonometric turning candidates; it can reduce over-approximation relative to IFK, but does not certify coupled extrema. An analytical extrema source solves low-dimensional sinusoidal boundary subproblems, reducing one- and two-joint boundary cases to structured root search. It becomes conservative only with an interval-closure wrapper; the Analytical row in Table VII is otherwise a comparison reference. Without such a wrapper, critical-sample and analytical endpoint sources are filters rather than certificate evidence. Any path using either source is accepted only after fixed-step validation unless wrapped by an independent conservative certificate. During LECT tree descent, transient prefix transforms are reused so unaffected kinematic-chain prefixes are not recomputed. This affects runtime materialization only; it is not persistent scene state. The upper panel of Fig. 1 summarizes this endpoint-to-link conversion before the scene check accepts a box.

C. C-space OBB Endpoint Envelopes

Axis-aligned boxes remain the global forest primitive. Bridge and repair motions introduce local joint correlations. For a bridge transition to carry a certificate, RBF validates a local typed C-space OBB-corridor region with the finite-cover test below, while leaving the dyadic forest unchanged. Without a successful finite cover, the same transition remains only a typed candidate that requires fixed-step validation. OBBs are defined in scaled joint coordinates. Let q_c be the local corridor center, and let $S \in \mathbb{R}^{n \times n}$ be a diagonal joint-scaling matrix with positive diagonal entries, defining the local metric. Let $m \leq n$ be the local dimension, $U \in \mathbb{R}^{n \times m}$ have scaled-metric orthonormal columns, and $h \in \mathbb{R}_{\geq 0}^m$ be the half-width vector. With the local cube $\Xi = \{\xi \in \mathbb{R}^m \mid -1 \leq \xi_i \leq 1, i = 1, \dots, m\}$ and $A = S^{-1}U \text{diag}(h) \in \mathbb{R}^{n \times m}$, the local region is

$$R_{\text{obb}} = \{q_c + A\xi \mid \xi \in \Xi\}.$$

The object R_{obb} is a configuration-space region, not a workspace OBB: the certificate comes from endpoint and link envelopes over R_{obb} , not from the fitted orientation.

a) *Cover certificate*: The conservative cover test tiles the local cube into subcubes X_τ , where τ indexes the cover tiles; for each $X_\tau \subseteq \Xi$, it forms the joint-space AABB hull

$$I_\tau = \text{bbox}_q(q_c + AX_\tau),$$

where $\text{bbox}_q(\cdot)$ denotes the joint-space AABB hull. If every I_τ passes the standard conservative link-envelope scene test, then

$$R_{\text{obb}} \subseteq \bigcup_{\tau} I_\tau \subseteq \mathcal{C}_{\text{free}},$$

where $\mathcal{C}_{\text{free}}$ is the collision-free configuration set under Assumption 1. When this test succeeds, the stored OBB certificate is a finite cover of axis-aligned box certificates.

b) *Correlated endpoint envelope*: A tighter source keeps the local OBB variables ξ correlated through the forward-kinematics computation. For each joint coordinate $j = 1, \dots, n$,

$$q_j = q_{c,j} + a_j^\top \xi, \quad \xi_i \in [-1, 1], \quad i = 1, \dots, m,$$

where $a_j^\top$ is the j -th row of A . Affine-arithmetic propagation then produces endpoint records with center c_k , generators $g_{k,i}$, unit noise variables $\epsilon_i \in [-1, 1]$, and a coordinatewise nonnegative remainder vector ρ_k :

$$E_k(R_{\text{obb}}) = c_k + \sum_i g_{k,i} \epsilon_i \oplus [-\rho_k, \rho_k].$$

The sum runs over the affine-noise generators in the endpoint record. Equivalently, with $p_k(q_c)$ the endpoint position and $J_k(q_c)$ its Jacobian at the corridor center,

$$E_k(R_{\text{obb}}) = p_k(q_c) + J_k(q_c)A\Xi \oplus [-\rho_k, \rho_k],$$

where $\mu \in \{1, 2, 3\}$ indexes workspace coordinates, the sum is over local coordinates $a, b = 1, \dots, m$, and each coordinate remainder is bounded conservatively, for example by

$$\rho_{k,\mu} \geq \frac{1}{2} \sum_{a,b} |(A^\top H_{k,\mu} A)_{ab}|,$$

with $H_{k,\mu}$ an interval or otherwise conservative Hessian bound over R_{obb} . If these remainders are sampled rather than outward bounded, the OBB source is a non-conservative filter and not a certificate.

OBB endpoint records feed the same endpoint-to-link construction as axis-aligned C-space boxes. These records refine a local typed edge; they do not add OBB nodes to the global axis-aligned forest. For a workspace support direction $d \in \mathbb{R}^3$, the support function of a zonotope-plus-box endpoint record is

$$h_{E_k}(d) = d^\top c_k + \sum_i |d^\top g_{k,i}| + \sum_{\mu=1}^3 |d_\mu| \rho_{k,\mu}.$$

Here the sum over i spans the affine-noise generators in the endpoint record, and μ indexes workspace coordinates. Because E_a and E_b enclose the endpoint trajectories, a conservative support bound for the zero-radius link centerline is

$$\bar{h}_\ell^\circ(d) = \max\{h_{E_a}(d), h_{E_b}(d)\},$$

and radius lifting gives the rounded-link support bound

$$\bar{h}_\ell(d) = \bar{h}_\ell^\circ(d) + r_\ell \|d\|_2.$$

Coordinate-axis queries of $\bar{h}_\ell$ provide an AABB broadphase, while the same support record feeds the SupportHull/GJK narrow phase. An OBB-corridor edge supports a certificate only when its endpoint bounds, finite-cover test, and separation tests are conservative; otherwise it is a typed transition that requires fixed-step validation. OBB-corridor records belong to the scene-local typed graph, not the persistent dyadic LECT tree.

D. Link-Envelope Representations

Endpoint records can be exported to multiple link representations without changing the upstream kinematic source. For a serial-chain capsule link ℓ whose proximal and distal endpoint records are $E_{\ell-1}(I)$ and $E_\ell(I)$, $B_\ell(I) = B_\ell^\circ(I) \oplus \mathcal{R}(r_\ell)$, with $B_\ell^\circ(I)$ bounding the centerline sweep and $\mathcal{R}(r_\ell)$ denoting the selected radius-lift set for link radius r_ℓ .

a) *AABB envelope*: AABB envelopes wrap the two endpoint envelopes in one box and pad by the link radius:

$$B_\ell(I) = \text{bbox}(E_{\ell-1}(I) \cup E_\ell(I)) \oplus \mathbb{B}_\infty(r_\ell).$$

Here $\text{bbox}(\cdot)$ denotes the workspace AABB hull. Segment subdivision may retain an explicit union of padded sub-envelopes for more selective filtering; collapsing that union back to one box would lose the tighter bound.

b) *Fixed-direction slab envelope*: A k -direction discrete oriented polytope (k -DOP) record stores lower and upper support values along a fixed direction set. Endpoint boxes are projected onto those directions, padded by the link radius, and merged over retained segment pieces. In the runtime pipeline, these slab tests form a low-cost intermediate filter: after the padded AABB broadphase, non-axis directions can reject close obstacle candidates before GJK. Slab tests are filters unless their endpoint source and separation tests are conservative.

c) *SupportHull envelope*: SupportHull keeps the two endpoint AABBs and link radius as a compact support record. For direction d , the support point is selected from the proximal or distal endpoint box, and obstacle refinement uses a fixed-size GJK narrow phase with outward obstacle inflation. The support record preserves more directional structure than a single AABB while keeping the cached record compact. When present, fixed-direction slab records act only as an intermediate filter between the AABB broadphase and SupportHull/GJK refinement; they do not change the certificate condition, which still depends on conservative endpoint evidence and outward scene tests.

IV. LIFELONG ENVELOPE CACHE TREE

LECT makes warm scene construction inexpensive without turning old obstacle labels into new certificates. Its invariant is evidence/label separation: it stores scene-independent endpoint and link-envelope evidence keyed by robot kinematics and joint intervals, while obstacle tests, box labels, graph connectivity, and query outcomes remain scene-local. Every replayed envelope is therefore checked against the current obstacle set.

A. Cache Key and Split Policy

LECT is a dynamically grown binary k -d tree over $\mathcal{Q}$. A node stores its joint interval, split metadata, cached endpoint and derived link-envelope evidence, and runtime residency state. The central invariant is that evidence is *kinematics-keyed* rather than *scene-keyed*: replay imports geometry bounds, not scene-validation labels or graph edges.

a) *Evidence/label separation*: An envelope record is replay-compatible only when the robot model, endpoint source, envelope representation, split descriptor, link radii, and active-link set match the current build. The split policy is part of this descriptor, so alternative policies define distinct cache descriptors. A descriptor match permits reuse of geometry bounds only; it never imports scene-validity labels. For a parent interval I , let I_d^- and I_d^+ be the two children obtained by splitting joint dimension d at its midpoint. Define the cumulative downstream envelope volume as

$$\mathcal{V}(I) := \sum_{\ell} \text{vol}(B_\ell(I)).$$

Here $\text{vol}(\cdot)$ denotes workspace volume. One implemented schedule chooses depth-wise split dimensions that minimize the worst-child cumulative envelope volume,

$$d^* = \arg \min_d \max \{ \mathcal{V}(I_d^-), \mathcal{V}(I_d^+) \}.$$

Depth-synchronous schedules keep replay deterministic; if the scheduled dimension is unavailable at a node, the runtime falls back to the widest valid dimension. Candidate dimensions are filtered by active-link reachability: splitting a joint that cannot affect any retained endpoint cannot tighten the downstream envelopes and is skipped for that descriptor.

b) *Sparse interval staging*: Planning cells need not materialize every ancestor. Each cell stores its exact joint interval, split counts, and descriptor fingerprint, which identifies the endpoint source, envelope representation, split policy, link radii, and active-link set used to interpret replay records. If no exact replay record is present, the runtime materializes the interval under that descriptor before validation.

B. Runtime Use and Validation Semantics

LECT serves local box materialization; graph search uses the scene forest. Given a joint interval, the runtime replays or materializes endpoint evidence, derives the requested link representation, and applies the current scene-validation policy. In the evaluated pipeline, AABB broadphase precedes SupportHull/GJK refinement. Parent or child records can accelerate materialization, but replayed evidence alone is not a collision-free certificate. When the required child records are present, their hulls can tighten an ancestor envelope; the reconstructed record remains representation-dependent and must still pass the current scene-validation policy. Directional records are staged filters: a joint interval first passes the padded

Panel legends: end-effector (EE), configuration-space obstacle (C-obstacle), waypoint count (wpt), and θ_1 and θ_2 denote schematic joint coordinates.

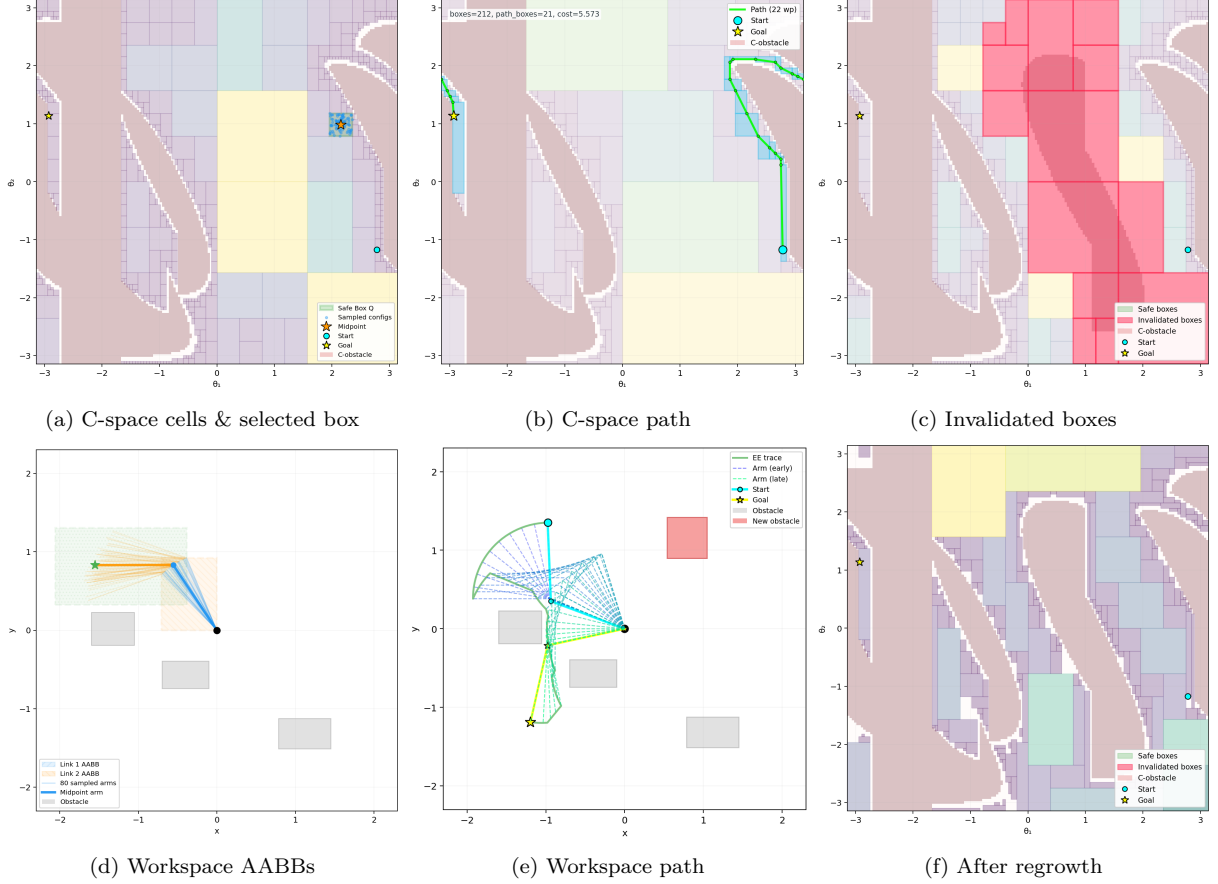


Fig. 2. Two-link schematic of the RBF build, query, and update cycle. A C-space box maps to workspace axis-aligned bounding boxes (AABBs); a box-corridor supplies the path; obstacle insertion triggers local refill. Panels show schematic logic, not experiment geometry; in (e), traces and arm labels denote schematic poses.

AABB broadphase, and only the remaining close obstacle candidates reach slab or SupportHull/GJK refinement. If no exact replay record exists, the interval is materialized on demand and then handled by the same validation path, so a cache miss changes cost, not semantics or validation responsibility.

C. Storage and Warm-Build Semantics

Persistence is selective. Endpoint records and derived link-envelope records are the reusable evidence units. IFK+AABB records can be regenerated on demand. Non-AABB envelope records are stored only when their additional rejection power offsets lookup and storage cost. The prewarm pass fills target leaf envelope records and derives ancestor hulls from child evidence. This avoids recomputing direct forward kinematics independently at every ancestor depth. In a warm build, a cache hit reconstructs the requested link envelopes; the scene builder then checks those envelopes under the current scene-validation policy. Warm builds still choose and grow scene-specific boxes; no persisted box label is imported as a certificate. The Shelf replay used in the experiments is read-only during timing, so the replay store does not accumulate new evidence during those measurements. Random-scene

cache growth is treated separately as part of the per-robot profile.

V. RAPIDBOXFOREST CONSTRUCTION

This section turns envelope validation and LECT replay into a query-agnostic scene structure. Fig. 2 illustrates the planning layer: RBF builds scene-specific policy-accepted C-space boxes and stores their adjacency in a typed corridor graph. For a fixed obstacle set, later queries use box membership and graph search instead of rebuilding the cover. LECT remains a separate evidence store, and each scene box is checked against the current obstacles.

A box is an axis-aligned joint interval $B = [l^{(1)}, u^{(1)}] \times \dots \times [l^{(n)}, u^{(n)}] \subseteq \mathcal{Q}$. A *policy-accepted box* has passed the configured scene-validation policy. Conservative link-envelope disjointness makes acceptance a collision-free certificate; a non-conservative filter makes the box only a planning region, so any path segment using it remains validation-required. Rejected cells are excluded from the forest, while selected blocked or inconclusive domains are retained for local refinement or later update.

A. Construction Objective

The construction objective is a budgeted reusable cover, not an exhaustive decomposition of $\mathcal{C}_{\text{free}}$. Given the joint-

Algorithm 1 Local box materialization around a seed.

Input: Seed q , LECT tree $\mathcal{T}_L$, obstacle set $\mathcal{O}$, validation policy $V_{\mathcal{O}}$, descriptor δ , depth cap $D_{\max}$
Output: Accepted interval with status, or failure with retained domain

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1:  $I \leftarrow$  root interval of  $\mathcal{T}_L$  that contains  $q$ ;  $d \leftarrow 0$ 
2: while  $d \leq D_{\max}$  do
3:    $R \leftarrow \text{ReplayOrMaterialize}(\mathcal{T}_L, I, \delta)$ 
4:   derive active-link envelopes  $\{B_{\ell}(I)\}$  from endpoint record  $R$ 
5:    $s \leftarrow V_{\mathcal{O}}(\{B_{\ell}(I)\}, \mathcal{O})$ 
6:    $s$  is free, blocked, or inconclusive under the active policy
7:   if  $s$  is free under a conservative policy then
8:     return  $(I, \text{certificate-backed})$ 
9:   else if  $s$  is free only under a filter policy then
10:    return  $(I, \text{validation-required})$ 
11:   end if
12:   if  $d = D_{\max}$  then
13:     return failure with retained domain  $(I, s)$ 
14:   end if
15:   choose split dimension from  $\delta$ ; fall back to widest active dimension
16:    $I \leftarrow$  child of  $I$  that contains  $q$ ;  $d \leftarrow d + 1$ 
17: end while
18: return failure

```

limit box $\mathcal{Q}$ and obstacle set $\mathcal{O}$, let $V_{\mathcal{O}}(B)$ denote the configured scene-validation predicate for a box B . The build stage constructs a forest $\mathcal{F}$ of N retained boxes satisfying

$$\mathcal{F} = \{B_1, \dots, B_N\},$$

$$B_i \subseteq \mathcal{Q}, \quad V_{\mathcal{O}}(B_i), \quad i = 1, \dots, N.$$

The reusable output is $(\mathcal{F}, G)$, where G is the typed corridor graph storing adjacency and auxiliary transitions; the box budget targets anchors, frontiers, and connectors.

For two boxes

$$B_i = \prod_{d=1}^n [l_i^{(d)}, u_i^{(d)}], \quad B_j = \prod_{d=1}^n [l_j^{(d)}, u_j^{(d)}],$$

let $\varepsilon_c \geq 0$ denote the connection tolerance. RBF treats the boxes as connected when the closed intervals in every coordinate overlap or have a gap no larger than ε_c :

$$\max\{l_i^{(d)}, l_j^{(d)}\} \leq \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \quad d = 1, \dots, n.$$

Only exact overlap or contact directly yields a contained box-corridor. Tolerance-gap and explicit segment edges are graph-search aids, not certificates, unless they are converted into conservative box-corridor or finite-cover OBB-corridor regions; otherwise they require fixed-step validation.

B. Build Pipeline

The build proceeds through four stages. First, local materialization descends the LECT tree around a seed and returns the first interval accepted by the configured scene policy. Second, a priority leaf sweep classifies cells near anchors, frontiers, and uncovered query-relevant domains. Third, restricted refinement revisits retained blocked or inconclusive domains under a separate budget. Finally, consolidation merges or removes boxes only after revalidating the resulting box. Algorithms 1 and 2 make these decisions explicit; all stages use the same materialize-validate-connect pattern, so later graph search never relies on an unvalidated scene label.

Algorithm 2 Adaptive leaf sweep and local refinement.

Input: Seed/anchor set $\mathcal{S}$, LECT tree $\mathcal{T}_L$, obstacle set $\mathcal{O}$, validation policy $V_{\mathcal{O}}$, sweep and refinement budgets
Output: Scene forest $\mathcal{F}$, typed graph G , retained domains $\mathcal{C}$

```

1: initialize priority queue  $\mathcal{P}$  using the fixed priority policy
2:  $\mathcal{F} \leftarrow \emptyset$ ;  $G \leftarrow \emptyset$ ;  $\mathcal{C} \leftarrow \emptyset$ 
3: while  $\mathcal{P}$  is nonempty and sweep budget remains do
4:   pop the highest-priority cell or seed domain from  $\mathcal{P}$ 
5:   call Algorithm 1; if the domain is fixed, materialize that exact cell
6:   if an accepted box  $B$  is returned then
7:     insert  $B$  into  $\mathcal{F}$  and connect overlap/contact neighbors in  $G$ 
8:     propose connector, gap, segment, OBB, and repair transitions within the fixed profile budget
9:     convert eligible transitions to conservative corridors; insert each edge in  $G$  with the status in Table II
10:    enqueue adjacent frontier domains not already covered by  $\mathcal{F}$ 
11:   else
12:     retain the blocked or inconclusive domain in  $\mathcal{C}$ 
13:     split by the descriptor policy and re-enqueue children while the refinement budget allows
14:   end if
15: end while
16: for all retained domains selected from  $\mathcal{C}$  do
17:   refine locally; insert only boxes that pass  $V_{\mathcal{O}}$  and remain in the domain
18: end for
19: for all candidate merges or redundant boxes do
20:   replace boxes only if the merged or retained box passes the same validation rule
21: end for
22: return  $\mathcal{F}, G, \mathcal{C}$ 

```

a) *Controls:* Validation semantics and timings are governed by the build budget, LECT descriptor, endpoint and link representation, admitted edge types, and validation resolution. The fixed profile uses saved anchors, query endpoints, frontier domains, and repair witnesses as seeds; priority favors anchor and connector coverage before uncovered frontier volume, with deterministic tie-breaking. Splits follow the LECT descriptor schedule, profile budgets are fixed before evaluation, and candidate merges are kept only after revalidation. Materialization failures remain deferred domains for restricted refinement; frontier and repair witnesses follow the status rules in Table II. In Algorithms 1 and 2, $\mathcal{T}_L$, $D_{\max}$, and $\mathcal{C}$ denote the LECT tree, the descent depth cap, and the retained blocked or inconclusive collision-domain cache.

Exact-cell materialization means replaying or computing evidence for the retained cell itself rather than descending from a seed-containing ancestor. This keeps edit, refill, and repair domains aligned with the scene-cache labels that caused the revisit.

C. Corridor Query and Certificate

The graph is typed because search connectivity and certificate-carrying connectivity need not coincide. After consolidation, G has policy-accepted vertices and edge types $E_{\cap}$ (overlap/contact), E_{ε} (tolerance gap), E_s (segment), E_o (finite-cover OBB-corridor), and E_{π} (repair witness). As summarized in Table II, overlap/contact edges and finite-cover OBB-corridor edges carry certificates only when backed by conservative region checks. Repair-witness

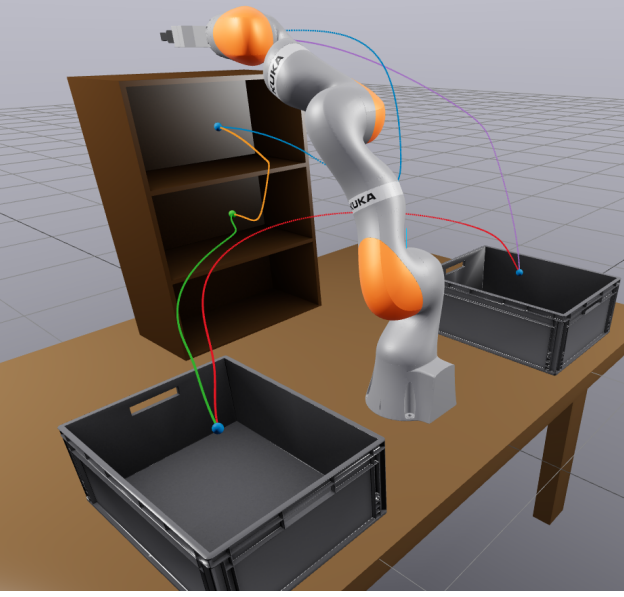


Fig. 3. Canonical Marcucci Shelf+KUKA LBR iiwa (Shelf+IIWA) scene with saved anchors: approach-side (AS), target-shelf (TS), center-shelf (CS), left-bin (LB), and right-bin (RB).

edges carry certificates only after conversion into a conservative box-corridor or OBB-corridor representation. Certificate status is determined by the backing conservative region check, not by the edge label alone.

Given start and goal configurations, the query locates containing boxes or configured anchor boxes and searches G . The certificate applies only to returned paths contained in conservative box regions or finite-cover OBB-corridor regions. Lemma 1 states the box-region case; Section III-C gives the finite-cover certificate for OBB-corridor edges. All other returns follow the validation-required semantics in Table II.

Lemma 1 (Box-corridor certificate). *Let $\mathcal{F}$ be a forest whose boxes have each passed conservative link-envelope validation against obstacle set $\mathcal{O}$. Let $R_{\mathcal{F}} := \bigcup_{B \in \mathcal{F}} B$ denote the represented box union. If a continuous path $\gamma : [0, 1] \rightarrow \mathcal{Q}$ satisfies $\gamma([0, 1]) \subseteq R_{\mathcal{F}}$, then every configuration on γ is collision-free with respect to the collision model in Assumption 1.*

Proof. For any $s \in [0, 1]$, containment gives a conservatively validated box $B \in \mathcal{F}$ with $\gamma(s) \in B$. By conservative link-envelope validation and Corollary 1, the robot geometry at $\gamma(s)$ is disjoint from $\mathcal{O}$. Since s was arbitrary, the whole path is collision-free. $\square$

Certificate interpretation. The lemma certifies collision-freedom for paths contained in conservatively validated regions. It does not imply coverage or completeness: finite depth, axis-aligned boxes, envelope overapproximation, and bridge or repair limits can miss paths. Returns that leave certified regions, include residual non-corridor segments, are post-processed, or traverse boxes admitted by non-conservative filters follow the validation-required semantics in Table II.

VI. EXPERIMENTS

The evaluation is organized as an evidence chain. Q1 tests the intended warm reuse mechanism on the Shelf+KUKA LBR iiwa (Shelf+IIWA) scene. Q2 places that fixed profile against roadmap, sampling, and convex-region baselines. Q3 checks whether the same reusable-forest behavior holds across saved random scenes and three robot models. Q4 explains the mechanism choices behind the default profile, and Q5 isolates local edit maintenance.

a) Protocol: Timings were measured on Ubuntu 22.04 with an Intel i5-12600KF central processing unit (CPU) and 31 gibibytes (GiB) of random-access memory (RAM). All RBF, LECT, and OMPL comparison rows use the common 8-thread wall-clock policy. The generated tables and figures store data sources, simplification settings, checksums, and integrity metadata.

For planning tables, Build is reusable scene-construction time before the timed queries. Online/q is planner online time after a reusable build; it excludes final simplification and the post-run validation pass. Success nevertheless requires fixed-step validation with a 0.01 rad segment step unless a row is explicitly marked conservative. Thus claims using Online/q are planner-time claims, not total accepted-path latency; the validation pass is a common acceptance audit rather than an input to the timing ranks. If a row or supplementary artifact reports validation or simplification costs, those fields must be added explicitly before making a total-latency claim; otherwise compact main-table timings should be read strictly as planner-time comparisons. Entries report median $[Q_1, Q_3]$ unless the caption states otherwise. Amortized quantities of the form Build/ K + Online/q use K as a stated reuse horizon, not a success count. Path-length ratios divide the validated returned path by the same-query empirical reference length; L_{ref} denotes that empirical reference, not an optimality certificate.

Planning rows use the collision model in Assumption 1: active rounded links are checked against $\mathcal{O}$. Self-collision, attachments, grippers, and task-specific forbidden regions require additional policy checks before a certificate is claimed.

b) Cache and profile accounting: Warm-cache rows include compatible LECT lookup and evidence reads, but exclude snapshot creation/loading, resident memory, and offline setup amortization. The d23 label identifies a depth-23 LECT cache: 16,777,215 records, 12.75 GiB on disk, and 958.079 s prewarm time. Bridge and repair witnesses follow Table II: after conservative box-corridor or finite-cover OBB-corridor conversion they are certificate-backed; otherwise they remain validation-required transitions. All profile budgets are fixed before producing the Shelf and random-scene summaries; the b100 Shelf profile and random-scene OBB-corridor profiles use these fixed budgets unchanged across saved seeds and robot/difficulty rows, and no row is tuned after observing validation outcomes.

TABLE III

EVALUATION MAP. EACH QUESTION IS TIED TO ONE WRITING CLAIM AND ONE ACCOUNTING BOUNDARY SO THE EXPERIMENTAL NARRATIVE STAYS ALIGNED WITH TABLE II.

Question	Writing claim	Main evidence	Boundary emphasized
Q1 Shelf replay	LECT replay lowers reusable scene-build cost.	Fixed b100 Shelf profile, d23 cache, five-anchor cycle.	Replay imports geometry evidence only; success requires the same final validation.
Q2 Shelf context	RBF occupies a low-build middle point.	PRM, RRT-Connect, BIT*, and IRIS/GCS on the same saved cycle.	Build/5 + Online/q is the reuse-horizon display, not an optimality claim.
Q3 Random scenes	Reusable forests generalize across robots and scene difficulty.	Three-robot medium/hard catalogs.	RBF is not claimed as fastest single-query planning.
Q4 Mechanisms	Endpoint and link-envelope choices explain the default profile.	Endpoint-source and narrow-phase mechanism tables.	Mechanism rows are not end-to-end planning rows.
Q5 Edits	Local scene-cache maintenance is isolated from warm rebuild cost.	Obstacle insertion/removal maintenance timing.	Maintenance-only timing excludes post-edit query success.

TABLE IV

SHELF+IIWA RBF PROFILE AND MECHANISM ABLATIONS.

Case	Build	Online/q	Amort. 5q	L/L^*
RBF baseline	0.171 [0.166, 0.174]	0.077 [0.072, 0.085]	0.111 [0.107, 0.117]	1.15 [1.08, 1.39]
Critical sample	0.279 [0.275, 0.283]	0.087 [0.078, 0.095]	0.145 [0.134, 0.151]	1.14 [1.09, 1.38]
Link AAB B	0.167 [0.164, 0.171]	0.083 [0.077, 0.086]	0.116 [0.111, 0.119]	1.19 [1.08, 1.48]
Live HIFK-5	2.770 [2.644, 3.003]	0.181 [0.139, 0.207]	0.739 [0.703, 0.777]	1.15 [1.08, 1.43]
SupportHull only	0.193 [0.188, 0.200]	0.087 [0.082, 0.095]	0.126 [0.120, 0.136]	1.15 [1.08, 1.39]

Notes: Median $[Q_1, Q_3]$, s. Build is reusable; Amort. 5q = Build/5 + Online/q; Online/q excludes simplification/validation. AAB B = axis-aligned bounding box; HIFK = hierarchical interval forward kinematics; SupportHull = support-function hull. Critical sample is non-certifying; L/L^* uses the same-query 0.01 rad reference.

Labels: IRIS/GCS = Iterative Regional Inflation + Graphs of Convex Sets; RBF b100 = fixed 100-box Shelf point; Amort. 5q = Build/5 + Online/q.

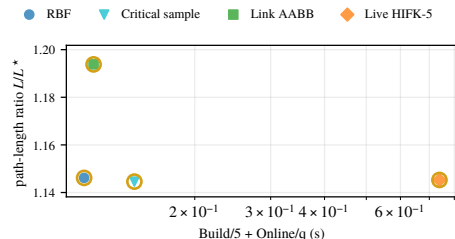


Fig. 4. Shelf+IIWA mechanism trade-off. Points plot Table IV Amort. 5q versus L/L_{ref} ; smaller ratios are shorter, and gold rings mark Table IV rows.

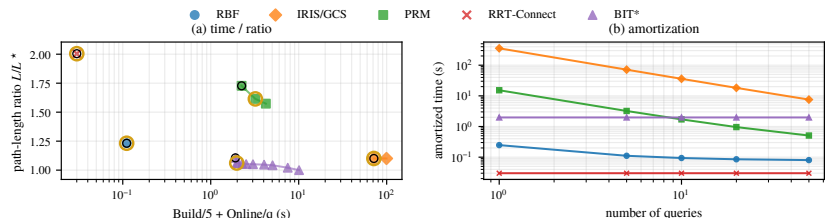


Fig. 5. Shelf+IIWA cross-algorithm trade-off. Black rings mark first successful check-points. All rows follow the common 8-thread execution policy. The horizontal coordinate is Build/5 + Online/q for reusable methods and Online/q for per-query rows with no reusable build; smaller L/L_{ref} is shorter. Right: amortized time per query versus reuse count.

TABLE V

SHELF+IIWA QUERY-VALIDATED CROSS-ALGORITHM COMPARISON.

Query	RBF b100 Build 0.171 s		IRIS/GCS Build 355.425 s		PRM Build 20.040 s		RRT-Connect No reusable build		BIT* No reusable build	
	T (s)	L/L^*	T (s)	L/L^*	T (s)	L/L^*	T (s)	L/L^*	T (s)	L/L^*
AS→TS	0.065 [0.064, 0.070]	1.40 [1.34, 1.65]	0.449 [0.449, 0.449]	1.44 [1.44, 1.44]	0.198 [0.147, 0.248]	1.61 [1.42, 1.73]	0.034 [0.024, 0.052]	2.27 [2.05, 2.72]	4.004 [4.002, 4.004]	1.06 [1.02, 1.08]
TS→CS	0.085 [0.082, 0.089]	1.13 [1.11, 1.14]	0.269 [0.269, 0.269]	1.26 [1.26, 1.26]	0.173 [0.118, 0.312]	2.99 [2.11, 3.22]	0.054 [0.021, 0.089]	3.98 [2.80, 5.09]	10.138 [10.004, 10.386]	1.09 [1.04, 1.11]
CS→LB	0.073 [0.070, 0.077]	1.47 [1.32, 1.82]	0.381 [0.381, 0.381]	1.07 [1.07, 1.07]	0.136 [0.120, 0.195]	1.49 [1.39, 2.09]	0.032 [0.027, 0.070]	2.20 [1.75, 2.51]	7.504 [7.504, 7.725]	1.03 [1.01, 1.06]
LB→RB	0.078 [0.075, 0.083]	1.09 [1.07, 1.22]	0.786 [0.786, 0.786]	1.07 [1.07, 1.07]	0.164 [0.091, 0.399]	1.18 [1.09, 1.27]	0.001 [0.001, 0.002]	1.29 [1.27, 1.32]	1.654 [1.654, 1.654]	1.02 [1.02, 1.03]
RB→AS	0.081 [0.077, 0.086]	1.04 [1.03, 1.05]	0.527 [0.527, 0.527]	1.01 [1.01, 1.01]	0.252 [0.096, 0.295]	1.23 [1.17, 1.24]	0.001 [0.001, 0.001]	1.07 [1.03, 1.10]	0.204 [0.204, 0.204]	1.02 [1.01, 1.03]

Notes: Median $[Q_1, Q_3]$, s. Header Build is reusable; T = Online/q excludes simplification/validation. L/L^* uses the same-query 0.01 rad reference. IRIS/GCS = Iterative Regional Inflation/Graphs of Convex Sets; PRM = probabilistic roadmap; RRT-Connect = bidirectional RRT; BIT* = Batch Informed Trees. Anchors as in Fig. 3.

A. Q1: Shelf+IIWA Replay and Fixed Profile

Q1 tests whether LECT replay lowers reusable build cost without changing path-acceptance responsibility.

Fig. 3 shows the Shelf+IIWA scene and fixed five-anchor query cycle with approach-side (AS), target-shelf (TS), center-shelf (CS), left-bin (LB), and right-bin (RB)

anchors; the executed order is AS-TS-CS-LB-RB-AS. Table IV and Fig. 4 report the fixed 100-box Shelf profile (b100) over seeds 0–7. The cycle is used for mechanism ablations and profile selection; saved random-scene catalogs below test broader start-goal variation.

The b100 RBF baseline uses SupportHull link envelopes replayed from the d23 cache: Build 0.171 s, Online/q 0.077 s, and five-query amortized time 0.111 s; Table IV gives quartiles. The Live HIFK-5 row keeps the same depth and validation budget but computes evidence live, giving a 2.770 s reusable build; the other rows replay d23 evidence. After OBB-corridor conversion, the 40 validated Shelf paths in Table IV contain no residual non-corridor segments. The representation rows isolate implementation choices: Link AABB changes Online/q from 0.077 s to 0.083 s while changing the median L/L_{ref} from 1.15 to 1.19; SupportHull-only raises the five-query amortized time from 0.111 s to 0.126 s.

The Shelf profile table follows the guarantee matrix: converted corridor segments carry certificates, and any remaining acceptance is audited by the fixed 0.01 rad validation step. In the reported fixed pipeline, all 40 paths pass that audit. A disabled-pipeline ablation with the query bridge, repair, shortcutting, and final simplification all turned off is included only as a full-pipeline consistency check; it should not be read as a single-stage failure diagnosis.

Takeaway. LECT replay reduces the reusable build by about $16.2\times$ relative to live HIFK-5 under the same fixed profile, while success remains anchored in the same path-acceptance rules in Table II.

B. Q2: Shelf Cross-Algorithm Context

Q2 tests the cross-algorithm context of the fixed RBF profile. The study applies the same five queries, 0.01 rad validation segment step, 0.010 s final-simplification budget, and 8-thread execution policy to RBF, IRIS/GCS, PRM, RRT-Connect, and BIT*. RBF uses the b100 profile and per-query Online/q; PRM and IRIS/GCS report reusable builds, while RRT-Connect and BIT* remain per-query rows with no reusable build term under the same thread budget. BIT* checkpoints are offline quality–time references, not online stopping rules.

Under these definitions, reusable builds are 0.171 s (RBF), 20.040 s (PRM), and 355.425 s (IRIS/GCS). The RBF build excludes d23 prewarm, snapshot loading costs, and resident memory. In the per-query T columns, RRT-Connect has lower T than RBF in all five Shelf queries but has higher L/L_{ref} in all five; BIT* has higher T than RBF but lower L/L_{ref} in all five. This timing–quality comparison is limited to the saved five-query cycle and its stated reusable build accounting.

Takeaway. On this fixed cycle, RBF is the low-build middle point: it avoids the large reusable builds of PRM and IRIS/GCS while returning shorter validated paths than RRT-Connect. The same data should not be read as a shortest-path or fastest-single-query claim, because BIT*

is shorter and RRT-Connect is faster per query in this saved cycle.

C. Q3: Random Multi-Robot Scenes

Q3 tests whether reusable box forests remain useful beyond the Shelf profile.

Dataset and pruning. Each saved random scene has ten pre-generated queries shared by every method. Across KUKA LBR iiwa, Universal Robots UR5, and Franka Emika Panda, the medium and hard catalogs over ten seeds contain 600 query instances before applying the fixed, method-independent pruning rule. For each robot and seed, medium and hard rows share the saved start–goal set; hard rows append a saved obstacle-prefix extension to the medium row, so difficulty changes are nested rather than regenerated. Before summarizing any method, the pruning rule removes cases where the direct start–goal segment already validates without planning. The retained post-pruning count, denoted N_{ret} , is reported with each robot/difficulty row in the random-summary table. All methods in a row share that retained set. The same table carries the IRIS/GCS full-success coverage field, so baseline coverage is visible without computing medians over method-specific survivors. The pruning changes summary statistics but not the saved catalogs, which retain ten queries per scene.

Compared methods and accounting. The trade-off plot uses the five-query amortized quantity $\text{Build}/5 + \text{Online}/q$ for rows with reusable builds and Online/q otherwise as a fixed display point; K is a reuse horizon, not a post-pruning retained-query count. Table VI and Fig. 6 summarize the operating points and corresponding checkpoint curves. RBF builds one query-agnostic forest per scene and reuses it for the saved query set; medians are computed over retained post-pruning queries. Random-scene RBF rows use LECT caches and anchors tied to each robot rather than Shelf-specific assets. Each RBF row uses the OBB-corridor profile and passes fixed-step validation on the summarized query set. In those rows, accepted paths use only converted finite-cover C-space OBB-corridor repair witnesses; no residual non-corridor segments remain. PRM is built once per scene, whereas RRT-Connect and BIT* are invoked independently for each query; all OMPL rows follow the common 8-thread execution policy specified in the protocol. IRIS/GCS did not satisfy the full validated-success criterion required for comparable random-scene median Online/q and path-length-ratio rows under the fixed saved-catalog protocol. The random-summary table therefore reports $N_{\text{full}}/6$, where N_{full} is the number of robot/difficulty rows satisfying that rule, rather than a median operating point; medians over a method-specific surviving subset are intentionally not reported. Compact random-scene rows without audit-cost fields are planner-time rows rather than total accepted-path latency rows.

Results. RRT-Connect has the lowest median Online/q in five of the six operating points; in Panda-Hard, RBF

TABLE VI
SAVED-CATALOG RANDOM-SCENE OPERATING POINTS.

Scenario	RBF			PRM			RRT-Connect		BIT*	
	Build	Online/q	L/L_q^*	Build	Online/q	L/L_q^*	Online/q	L/L_q^*	Online/q	L/L_q^*
IIWA-Medium	0.075	0.009	1.00	3.567	0.038	1.03	0.002	1.02	0.107	1.00
	[0.058, 0.083]	[0.007, 0.011]	[1.00, 1.03]	[3.007, 3.933]	[0.014, 0.070]	[1.02, 1.12]	[0.001, 0.002]	[1.01, 1.05]	[0.087, 0.154]	[1.00, 1.01]
IIWA-Hard	0.097	0.014	1.00	4.175	0.048	1.03	0.002	1.03	0.129	1.00
	[0.086, 0.105]	[0.011, 0.015]	[1.00, 1.03]	[3.408, 5.637]	[0.021, 0.102]	[1.02, 1.08]	[0.002, 0.003]	[1.00, 1.07]	[0.097, 0.186]	[1.00, 1.01]
UR5-Medium	0.262	0.075	1.05	3.124	0.323	1.02	0.023	1.23	0.134	1.01
	[0.254, 0.270]	[0.020, 0.086]	[1.00, 1.14]	[2.966, 3.418]	[0.085, 0.426]	[1.00, 1.12]	[0.002, 0.049]	[1.06, 1.54]	[0.101, 0.167]	[1.00, 1.08]
UR5-Hard	0.276	0.083	1.05	2.488	0.419	1.06	0.048	1.36	0.154	1.03
	[0.264, 0.284]	[0.043, 0.108]	[1.01, 1.14]	[2.325, 3.048]	[0.129, 0.437]	[1.01, 1.19]	[0.008, 0.090]	[1.14, 1.58]	[0.115, 0.171]	[1.00, 1.13]
Panda-Medium	0.250	0.028	1.05	3.431	0.066	1.06	0.005	1.22	0.116	1.00
	[0.218, 0.265]	[0.013, 0.039]	[1.02, 1.09]	[3.364, 4.180]	[0.038, 0.159]	[1.03, 1.08]	[0.002, 0.011]	[1.12, 1.35]	[0.092, 0.152]	[1.00, 1.00]
Panda-Hard	0.224	0.043	1.05	2.337	0.054	1.10	0.056	1.25	0.164	1.01
	[0.221, 0.237]	[0.021, 0.053]	[1.01, 1.16]	[1.627, 3.978]	[0.034, 0.126]	[1.05, 1.16]	[0.019, 0.136]	[1.14, 1.37]	[0.123, 0.214]	[1.00, 1.09]

Notes: Median $[Q_1, Q_3]$, s. IIWA/UR5/Panda = KUKA LBR iiwa/Universal Robots UR5/Franka Emika Panda. Direct successes are pruned; catalogs retain ten queries/scene. Build is reusable for RBF/PRM; RRT-Connect/BIT* are single-query. Online/q excludes simplification/validation; L/L_q^* uses the same-query 0.01 rad reference.

Accounting note: The columns N_{ret} and IRIS/GCS $N_{\text{full}}/6$ make pruning and baseline coverage visible. N_{ret} is the method-independent retained-query count after direct-segment pruning for each robot/difficulty row, and $N_{\text{full}}/6$ counts robot/difficulty rows satisfying the full validated-success rule. Medians over method-specific survivors are not reported.

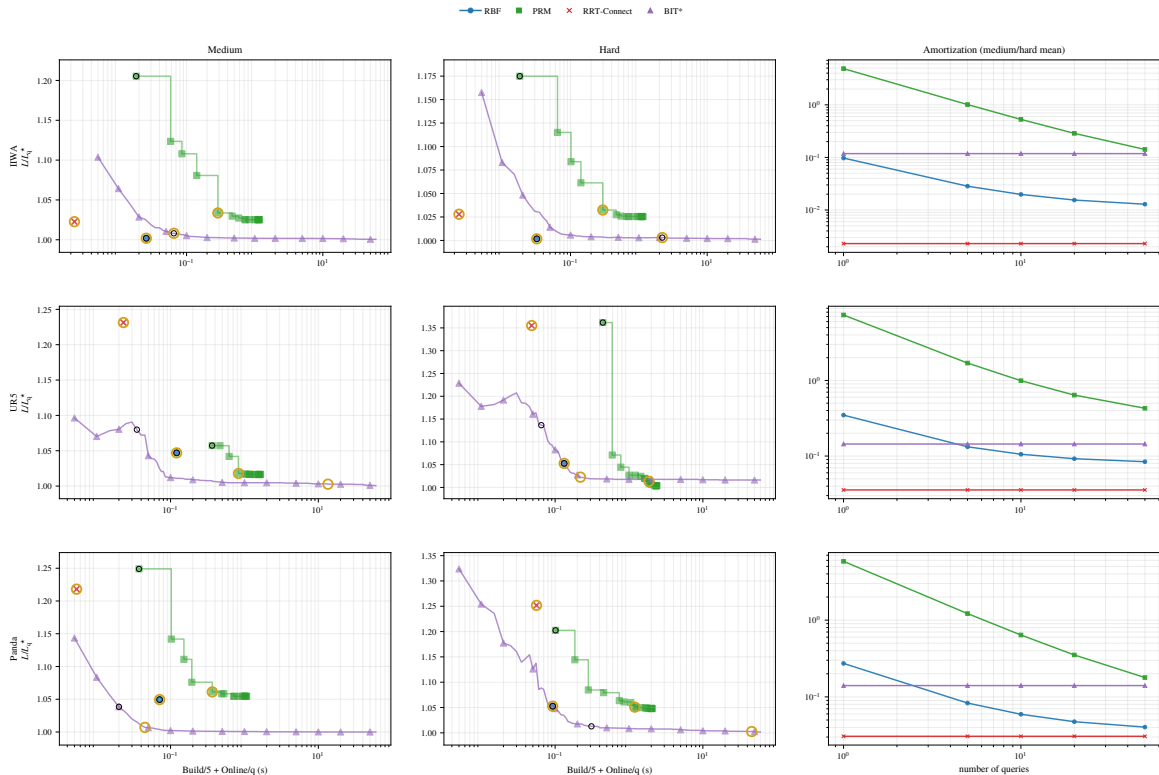


Fig. 6. Saved-catalog random-scene trade-off. Left/middle: fixed five-query point Build/5 + Online/q; right: reuse horizon K in Build/ K + Online/q. Online/q is per-query online time, and smaller path-length ratios are shorter. Rows are the three robot models. The summary table reports N_{ret} and IRIS/GCS $N_{\text{full}}/6$ coverage; no-build methods plot at Online/q under the common 8-thread policy. Rings mark first validated checkpoints (black) and later $\geq 1\%$ path-ratio gains (gold); PRM and BIT* use offline checkpoints.

has the lowest median Online/q. After fixed-step validation, the RRT-Connect median validated path-length ratio exceeds the RBF ratio in all six points. With scene-forest reuse, RBF has lower median Online/q than BIT* in all six points; $L/L_{\text{ref},q}$ matches BIT* to table precision in the two IIWA rows and is higher than BIT* in the four UR5/Panda rows.

Takeaway. The random catalogs support the intended role of RBF: reusable volumetric structure improves online time relative to BIT* while producing shorter validated paths than RRT-Connect. This is not a fastest-single-

query claim, because RRT-Connect remains faster in five of six median Online/q points.

D. Q4: Mechanism Studies Behind the Default Profile

Q4 links the default profile to mechanism measurements rather than end-to-end planning rows. Tables VII and VIII report joint-interval widths in radians. The endpoint table separates certificate-capable interval sources (IFK-AA and HIFK) from reference or filter rows; negative sampled-union gaps indicate finite-sample undercoverage, not certificate margins.

TABLE VII
ENDPOINT-ENVELOPE SOURCES OVER JOINT-INTERVAL SIDE LENGTHS.

Source	Volume ratio					Time (μ s)					Gap (mm)				
	0.02	0.10	0.20	0.30	0.50	0.02	0.10	0.20	0.30	0.50	0.02	0.10	0.20	0.30	0.50
IFK-AA	1.09	1.52	2.30	3.53	8.61	5.48	6.45	6.58	6.81	7.35	—	—	—	—	—
HIFK-3	1.04	1.22	1.51	1.89	3.12	25.94	29.16	27.21	29.27	29.38	—	—	—	—	—
HIFK-5	1.02	1.13	1.30	1.51	2.07	70.40	73.82	73.15	75.92	72.32	—	—	—	—	—
Critical sample	1.00	1.00	1.00	0.99	0.98	8.68	10.03	11.56	12.69	16.33	-0.05	-1.17	-4.53	-10.77	-30.97
Analytical	1.00	1.00	1.00	1.00	1.00	6429.96	6602.25	6800.10	6927.12	7096.01	0.00	0.00	0.00	0.00	-4.49
Monte Carlo	0.64	0.74	0.78	0.79	0.82	2251.74	11233.26	22352.95	33813.58	56835.57	-6.05	-32.68	-65.57	-88.87	-123.53

Notes: Widths are rad. Volume is normalized to Analytical; time is endpoint construction. IFK-AA = affine-arithmetic interval forward kinematics; HIFK = hierarchical interval forward kinematics. IFK-AA/HIFK certify; other rows are references. Gap = sampled-union signed gap; — = omitted.

TABLE VIII
LINK-ENVELOPE REPRESENTATIONS OVER JOINT-INTERVAL SIDE LENGTHS.

Envelope	Volume (m ³)						Build (μs)						Test (μs)					
	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50
Link AABB	0.0136	0.0201	0.0384	0.147	0.508	4.61	0.138	0.152	0.167	0.166	0.171	0.178	0.586	0.605	0.601	0.604	0.630	0.790
SupportHull	0.000359	0.00266	0.0143	0.106	0.449	4.54	0.205	0.214	0.218	0.219	0.217	0.250	12.520	12.850	12.267	12.223	12.659	13.026

Notes: Widths are rad; volume is mean workspace volume. AABB = axis-aligned bounding box; SupportHull = support-function hull. Test pools far/local probes; build excludes endpoint envelopes.

The link table explains the AABB-first narrow phase. SupportHull/GJK stores compact support records, but wide boxes can increase GJK calls after broadphase filtering. Link AABB is coarser, yet its build and hot-loop rejection costs are lower or comparable in the measured rows. The default profile therefore uses AABBs for broadphase rejection, interval evidence for conservative claims, and Table II for non-certificate-backed outputs.

Takeaway. The default profile uses certificate-capable interval evidence for box-level claims and a cheap AABB broadphase for hot-loop rejection; all remaining outputs inherit their status from Table II.

E. Q5: Dynamic-Update Maintenance Cost

Q5 isolates scene-cache maintenance after obstacle edits. Table IX reports cache-maintenance-only timing: its warm-build rows are scene-cache rebuilds, and no query bridge, connector, simplification, audit, or query timing is included.

TABLE IX
ADAPTIVE LEAF-SWEEP MAINTENANCE-ONLY TIMING.

Operation	Time (ms)	Speedup
Warm build (10 obstacles)	922.75 [897.57, 955.67]	—
Insert to 15 obstacles	6.06 [5.69, 6.54]	153.3–207.5×
Remove from 15 obstacles	0.35 [0.31, 1.65]	943.6–3354.0×
Warm build (15 obstacles)	958.79 [941.67, 1115.18]	—

Notes: Median $[Q_1, Q_3]$, ms. Maintenance only; speedup is relative to the 15-obstacle warm build.

This experiment therefore measures how quickly the scene cache updates its labels and local refill state after obstacle insertions or removals. It is not an end-to-end post-edit replanning benchmark; query success after an edit requires the same bridge, repair, simplification, and validation pipeline used by the planning studies above.

Takeaway. Maintenance-only rows show cache-update speedups of 153.3–207.5×

 for insertion and 943.6–3354.0×

VII. CONCLUSION

RBF is a reusable scene-construction layer for repeated manipulation queries. It makes the reused object volumetric and auditable: axis-aligned C-space boxes are the planning unit, endpoint-to-link swept-capsule bounds are the validation primitive, and LECT stores kinematics-keyed evidence without importing stale scene labels. The typed corridor graph records whether a returned path is backed by conservative boxes or finite-cover OBB corridors, or whether it follows the validation-required semantics in Table II.

The experiments support the intended middle-ground role. On Shelf+IIWA, d23 LECT replay reduces median reusable build time from 2.770 s for live HIFK-5 to 0.171 s under the same fixed profile. Across saved random-scene catalogs for three robot models, RBF is not the fastest per-query method in every case, but it returns shorter validated paths than RRT-Connect in all six operating points and lower median Online/q than BIT* in all six. Maintenance-only edit experiments show cache-update speedups of 153.3–207.5×

 for insertion and 943.6–3354.0×

for removal over a 15-obstacle warm rebuild.

a) Limitations: First, the certificate model covers active rounded links versus workspace obstacles; self-collision, attached objects, grippers, task regions, and other constraints require explicit policy and descriptor support. Second, Online/q excludes final simplification and post-run validation, and d23 prewarm, snapshot, and memory costs are outside the online protocol. Third, RBF is kinematic and collision-centric: dynamic feasibility, time-parameterized execution, and torque or contact constraints are outside the current claim. Fourth, RBF is not a complete planner: finite depth, axis-aligned overapproximation, and finite bridge/repair budgets can miss feasible paths. These extensions are compatible with the cache-and-policy design, but they require explicit descriptors and validation policies rather than following from the present certificates. These limits define the scope of the claim contract in Table II.

Within this scope, RBF shows that certificate-aware box-level reuse can provide a practical middle layer between roadmap reuse and convex-region optimization for repeated-query manipulation.

REFERENCES

- [1] R. Deits and R. Tedrake, “Computing large convex regions of obstacle-free space through semidefinite programming,” in *Algorithmic Foundations of Robotics XI*, 2015, pp. 109–124.
- [2] T. Marcucci, M. Petersen, D. von Wrangel, and R. Tedrake, “Motion planning around obstacles with convex optimization,” *Sci. Robot.*, vol. 8, no. 84, 2023.
- [3] T. Marcucci, J. Umenberger, P. Parrilo, and R. Tedrake, “Shortest paths in graphs of convex sets,” *SIAM J. Optim.*, vol. 34, no. 1, pp. 507–532, 2024.
- [4] S. M. LaValle, *Planning Algorithms*. Cambridge University Press, 2006.
- [5] —, “Rapidly-exploring random trees: A new tool for path planning,” Iowa State University, Tech. Rep. TR 98-11, 1998.
- [6] L. E. Kavraki, P. Švestka, J.-C. Latombe, and M. H. Overmars, “Probabilistic roadmaps for path planning in high-dimensional configuration spaces,” *IEEE Trans. Robot. Autom.*, vol. 12, no. 4, pp. 566–580, 1996.
- [7] S. Karaman and E. Frazzoli, “Sampling-based algorithms for optimal motion planning,” *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 846–894, 2011.
- [8] A. Amice, H. Dai, P. Werner, A. Zhang, and R. Tedrake, “Finding and optimizing certified, collision-free regions in configuration space for robot manipulators,” in *Algorithmic Foundations of Robotics XV*. Springer, 2022, pp. 328–348.
- [9] M. Petersen and R. Tedrake, “Growing convex collision-free regions in configuration space using nonlinear programming,” 2023. [Online]. Available: <https://arxiv.org/abs/2303.14737>
- [10] P. Werner, A. Amice, T. Marcucci, D. Rus, and R. Tedrake, “Approximating robot configuration spaces with few convex sets using clique covers of visibility graphs,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 10 359–10 365.
- [11] D. Blackmore and M. C. Leu, “Analysis of swept volume via Lie groups and differential equations,” *Int. J. Robot. Res.*, vol. 11, no. 6, pp. 516–537, 1992.
- [12] K. Abdel-Malek, J. Yang, D. Blackmore, and K. Joy, “Swept volumes: foundation, perspectives, and applications,” *Int. J. Shape Model.*, vol. 12, no. 1, pp. 87–127, 2006.
- [13] S. Gottschalk, M. C. Lin, and D. Manocha, “OBBTree: A hierarchical structure for rapid interference detection,” in *ACM SIGGRAPH*, 1996.
- [14] E. Larsen, S. Gottschalk, M. C. Lin, and D. Manocha, “Fast proximity queries with swept sphere volumes,” University of North Carolina at Chapel Hill, Tech. Rep. TR99-018, 1999.
- [15] E. G. Gilbert, D. W. Johnson, and S. S. Keerthi, “A fast procedure for computing the distance between complex objects in three-dimensional space,” *IEEE J. Robot. Autom.*, vol. 4, no. 2, pp. 193–203, 1988.
- [16] G. v. d. Bergen, “A fast and robust GJK implementation for collision detection of convex objects,” *J. Graph. Tools*, vol. 4, no. 2, pp. 7–25, 1999.
- [17] J. Pan, S. Chitta, and D. Manocha, “FCL: A general purpose library for collision and proximity queries,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2012, pp. 3859–3866.
- [18] R. E. Moore, R. B. Kearfott, and M. J. Cloud, *Introduction to Interval Analysis*. SIAM, 2009.
- [19] L. Jaulin, M. Kieffer, O. Didrit, and E. Walter, *Applied Interval Analysis*. Springer, 2001.
- [20] J. M. Snyder, “Interval analysis for computer graphics,” in *Proc. ACM SIGGRAPH*, 1992, pp. 121–130.
- [21] J.-P. Merlet, “Solving the forward kinematics of a Gough-type parallel manipulator with interval analysis,” *Int. J. Robot. Res.*, vol. 23, no. 3, pp. 221–235, 2004.
- [22] —, “Interval analysis for certified numerical solution of problems in robotics,” *Int. J. Appl. Math. Comput. Sci.*, vol. 19, no. 3, pp. 399–412, 2009.
- [23] S. Redon, Y. J. Kim, M. C. Lin, and D. Manocha, “Interactive and continuous collision detection for avatars in virtual environments,” in *IEEE Virtual Reality Conference*, 2004, pp. 117–124.
- [24] S. Redon, M. C. Lin, D. Manocha, and Y. J. Kim, “Fast continuous collision detection for articulated models,” *J. Comput. Inf. Sci. Eng.*, vol. 5, no. 2, pp. 126–137, 2005.
- [25] X. Zhang, S. Redon, M. Lee, and Y. J. Kim, “Continuous collision detection for articulated models using Taylor models and temporal culling,” *ACM Trans. Graph.*, vol. 26, no. 3, p. 15, 2007.
- [26] F. Schwarzer, M. Saha, and J.-C. Latombe, “Adaptive dynamic collision checking for single and multiple articulated robots in complex environments,” *IEEE Trans. Robot.*, vol. 21, no. 3, pp. 338–353, 2005.
- [27] J. A. Corrales, F. A. Candelas, and F. Torres, “Safe human–robot interaction based on dynamic sphere-swept line bounding volumes,” *Robot. Comput.-Integr. Manuf.*, vol. 27, no. 1, pp. 177–185, 2011.
- [28] R. Bohlin and L. E. Kavraki, “Path planning using lazy PRM,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2000, pp. 521–528.
- [29] C. M. Dellin and S. S. Srinivasa, “A unifying formalism for shortest path problems with expensive edge evaluations via lazy best-first search over paths with edge selectors,” in *International Conference on Automated Planning and Scheduling (ICAPS)*, 2016.
- [30] E. Schmerling, L. Janson, and M. Pavone, “Optimal sampling-based motion planning under differential constraints: The driftless case,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015.
- [31] D. Coleman, I. Sucan, S. Chitta, and N. Correll, “Reducing the barrier to entry of complex robotic software: A MoveIt! case study,” *J. Softw. Eng. Robot.*, vol. 5, no. 1, pp. 3–16, 2014.
- [32] T. Lozano-Pérez, “Spatial planning: A configuration space approach,” *IEEE Trans. Comput.*, vol. C-32, no. 2, pp. 108–120, 1983.
- [33] R. A. Brooks and T. Lozano-Pérez, “A subdivision algorithm in configuration space for findpath with rotation,” *IEEE Trans. Syst. Man Cybern.*, vol. SMC-15, no. 2, pp. 224–233, 1985.
- [34] J. J. Kuffner and S. M. LaValle, “RRT-Connect: An efficient approach to single-query path planning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2000, pp. 995–1001.
- [35] J. D. Gammell, S. S. Srinivasa, and T. D. Barfoot, “Batch Informed Trees (BIT*): Sampling-based optimal planning via the heuristically guided search of implicit random geometric graphs,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 3067–3074.
- [36] J. D. Gammell, T. D. Barfoot, and S. S. Srinivasa, “Informed sampling for asymptotically optimal path planning,” *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 966–984, 2018.
- [37] L. Janson, E. Schmerling, A. Clark, and M. Pavone, “Fast Marching Tree: A fast marching sampling-based method for optimal motion planning in many dimensions,” *Int. J. Robot. Res.*, vol. 34, no. 7, pp. 883–921, 2015.
- [38] I. A. Şucan, M. Moll, and L. E. Kavraki, “The Open Motion Planning Library,” *IEEE Robot. Autom. Mag.*, vol. 19, no. 4, pp. 72–82, 2012.
- [39] M. Moll, I. A. Şucan, and L. E. Kavraki, “Benchmarking motion planning algorithms: An extensible infrastructure for analysis and visualization,” *IEEE Robot. Autom. Mag.*, vol. 22, no. 3, pp. 96–102, 2015.
- [40] N. Ratliff, M. Zucker, J. A. Bagnell, and S. Srinivasa, “CHOMP: Gradient optimization techniques for efficient motion planning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2009, pp. 489–494.
- [41] M. Kalakrishnan, S. Chitta, E. Theodorou, P. Pastor, and S. Schaal, “STOMP: Stochastic trajectory optimization for motion planning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2011, pp. 4569–4574.
- [42] J. Schulman, Y. Duan, J. Ho, A. Lee, I. Awwal, H. Bradlow, J. Pan, S. Patil, K. Goldberg, and P. Abbeel, “Motion planning with sequential convex optimization and convex collision checking,” *Int. J. Robot. Res.*, vol. 33, no. 9, pp. 1251–1270, 2014.
- [43] M. Mukadam, J. Dong, X. Yan, F. Dellaert, and B. Boots, “Continuous-time Gaussian process motion planning via probabilistic inference,” *Int. J. Robot. Res.*, vol. 37, no. 11, pp. 1319–1340, 2018.